

Low Level Design – Cryptocurrency Volatility Prediction

1. Introduction

This document describes the Low Level Design (LLD) of the Crypto Volatility Prediction System. The purpose of this system is to analyze cryptocurrency market data and predict price volatility using machine learning techniques. The system processes historical cryptocurrency data, performs data preprocessing, creates useful features, trains a machine learning model, and generates predictions.

2. Data Collection Module

The data collection module is responsible for loading the cryptocurrency dataset. The dataset is stored in CSV format and contains features such as Open price, Close price, High price, Low price, Volume, and Timestamp. The dataset is loaded using the Python pandas library and stored in a dataframe for further analysis.

3. Data Preprocessing Module

This module prepares the raw dataset for machine learning. The preprocessing steps include: • Handling missing values • Removing infinite or invalid values • Converting data types if required • Normalizing or scaling numerical values These steps ensure the dataset is clean and suitable for model training.

4. Feature Engineering Module

Feature engineering helps create new useful variables from the existing dataset. Examples of features created in this project include:

- Price Change (Close – Open)
- Volatility measurement
- Rolling mean of price
- Percentage price change

These features help the machine learning model better understand patterns in cryptocurrency price movements.

5. Model Training Module

In this module, the dataset is split into training and testing sets using `train_test_split`. A machine learning algorithm such as Random Forest or Linear Regression is trained using the training dataset. The model learns patterns from historical cryptocurrency data to predict future volatility. The trained model is then saved for future predictions.

6. Prediction Module

The prediction module uses the trained machine learning model to generate predictions. The model takes processed features as input and predicts cryptocurrency volatility. These predictions help understand how unstable or stable the market may be.

7. Visualization Module

Data visualization is used to understand patterns in the dataset. Graphs are created using libraries such as Matplotlib and Seaborn. Examples of visualizations include:

- Price trend graphs
 - Volatility distribution plots
 - Correlation heatmaps
- These visualizations help in understanding relationships between different variables.

8. File Structure The project is organized into multiple folders for better code management.

Crypto-volatility-project

data/ → Dataset files

notebooks/ → Jupyter notebooks for EDA

src/ → Python scripts for preprocessing and training

models/ → Saved machine learning models

app/ → Application interface (if any)

reports/ → EDA report, HLD and LLD documents

9. Data Flow

Dataset → Data Preprocessing → Feature Engineering → Model Training → Prediction → Visualization

The dataset is first cleaned and processed. New features are generated from the data, then a machine learning model is trained. The trained model generates predictions, which are analyzed and visualized using graphs.

10. Conclusion

The Low Level Design explains how different modules of the system work together to build the cryptocurrency volatility prediction system. The modular structure ensures that each part of the system is easy to understand, maintain, and improve